

ZKM | Zentrum für Kunst und Medien

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Redesign of 2018 Version

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Audio subsystem concept.....	2
Audio generation and transmission	4
Tone generation.....	4
Low Pass Filter	4
Input divider	5
Amplifier	5
Loudspeaker	5
Audio detection	6
Microphone module.....	6
Audio analyzer	6
Prototype boards.....	7
Low Pass Filter	7
Microphone Module	7
Amplifier Module	8
Audio Analyzer Array	8

Audio subsystem concept

The audio subsystem comprises emitting and detection of single tones (aka frequencies). Each male or female is able to generate and detect single tones and each of them has his or her typical tone.

Figure 1 shows the transmitting part of the audio subsystem

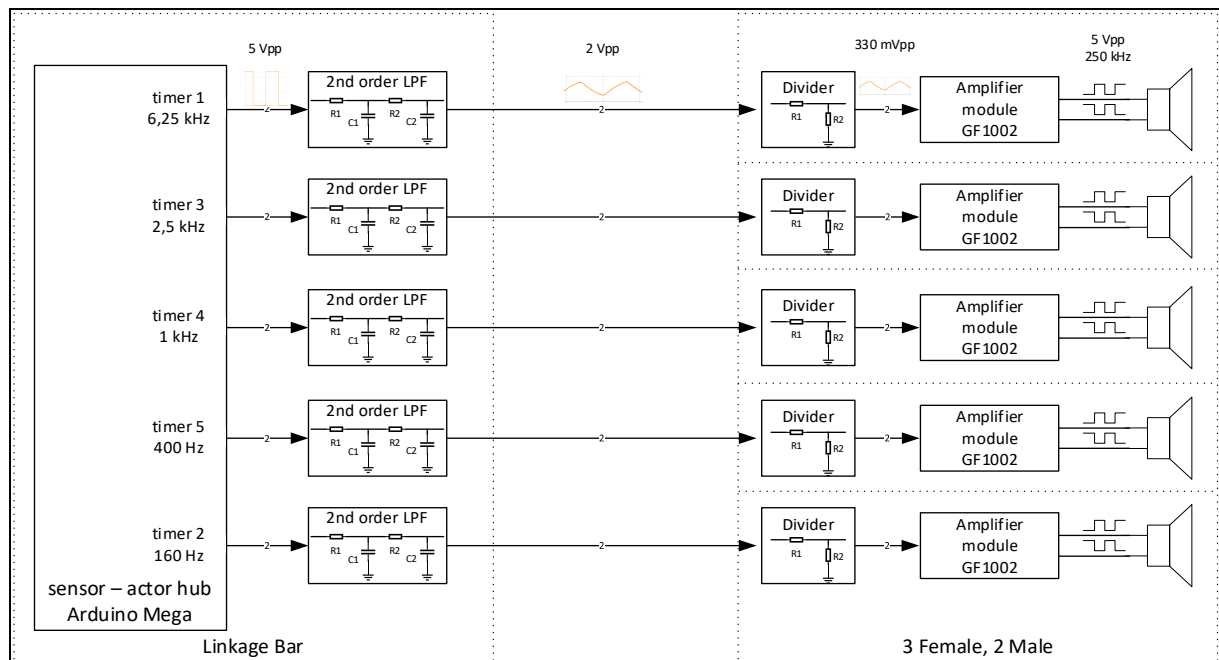


Figure 1: Audio subsystem - Audio generation

Figure 2 shows the receiving part of the audio subsystem

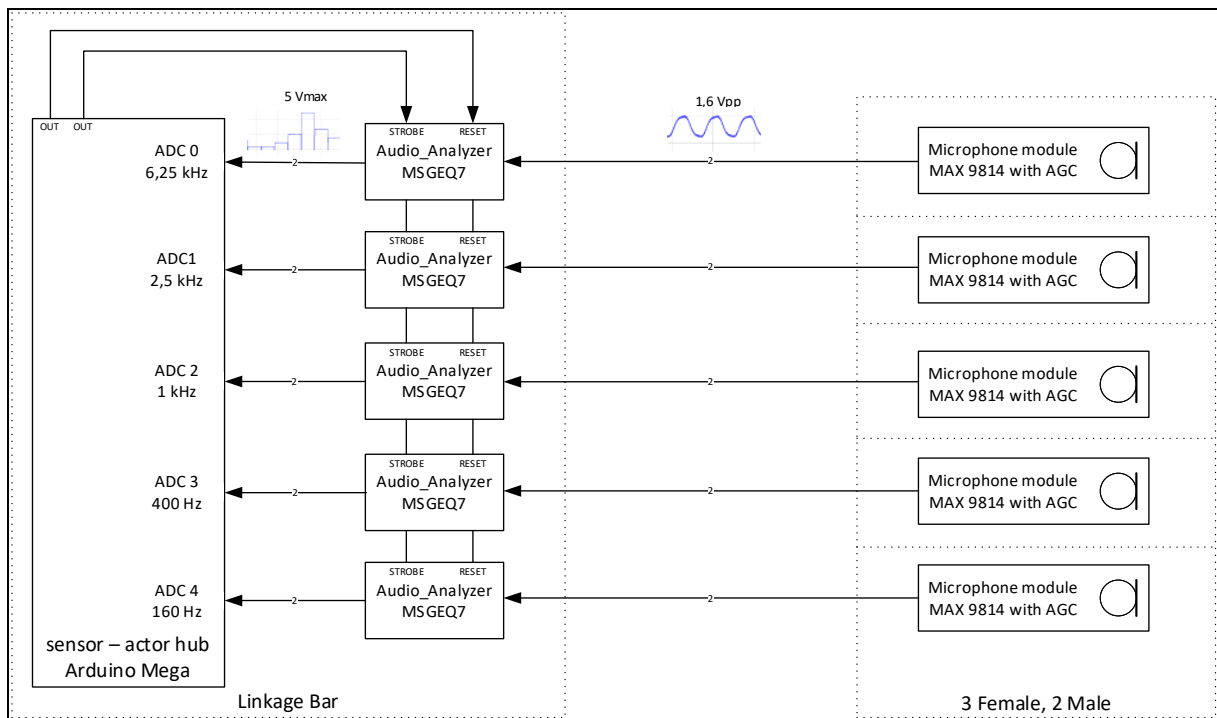


Figure 2: audio subsystem - Audio detection

In the following the structure of the audio subsystem and the design decisions are explained in more detail.

Audio generation and transmission

Tone generation

The audio analyzer module used for tone detection supports up to seven frequency bands: 63 Hz, 160 Hz, 400 Hz, 1KHz, 2.5 kHz, 6.25 kHz and 16 kHz.

Due to limitations in the frequency response of the components involved the 63 Hz and 16 kHz bands are of no choice. For example, typical electret microphones are specified in the frequency range of 100 Hz – 10000 Hz. And in addition, probably a significant amount of the audience will not even be able to hear the 16 kHz band anyway.

Since the ZKM concept uses a centralized sensor / actor hub the remaining five tones are generated by the Arduino Mega board. Timer 1, 3, 4, 5 (16-bit timers) and 2 (8-bit timer) are used for that purpose. With exception of the initialization, tone generation doesn't consume computing power since all timers run independently in CTC mode.

Low Pass Filter

Generating tones by means of counters makes life easier but has a big drawback: The result has a rectangular shape. Square waves can be thought of as the superposition of sine waves, starting with the fundamental frequency plus all odd harmonics. Especially for the first three frequency bands the bandpass filters of the audio analyzer are not able to completely filter out the first harmonic, see Table 1. This makes it much more difficult to safely detect the incoming fundamental frequency, especially in a noisy environment.

Square wave freq.	1 st harmonics	Neighbor band	Distance
160 Hz	480 Hz	400 Hz	80 Hz
400 Hz	1200 Hz	1000 Hz	200 Hz
1000 Hz	3000 Hz	2500 kHz	500 Hz
2500 Hz	7500 kHz	6250k Hz	1250 Hz
6250 Hz	18750 Hz	16000 Hz	2750 Hz

Table 1: Frequency bands and their first harmonics

The best solution would be generating sine waves only. This is not feasible with the centralized computing concept since it will block the CPU of the hub during tone generation. Therefore, it has been decided to use simple passive 2nd order low pass filter stages to damp the first and subsequent harmonics sufficiently enough to improve the quality of the frequency detection. Figure 3 shows the principal diagram of the second order filter.

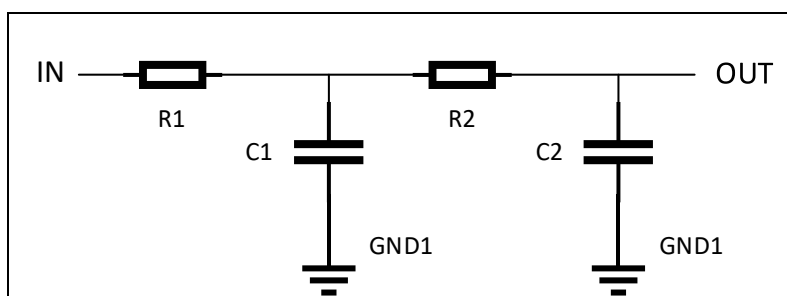


Figure 3: Passive 2nd order low pass filter

The output of the filter is still not a sine wave but measurements have shown the first harmonic damped by more than 20 dB = less than 10% to the fundamental frequency.

To simplify the design the two filter stages are identical, $R1 = R2$ and $C1 = C2$.

The following component values are used:

Band	R	C
6250 Hz	2K2	10nF
2500 Hz	1K8	47nF
1000 Hz	1K2	150nF
400Hz	2K	220nF
160 Hz	2K2	470nF

Table 2: Filter component values

It has been decided, while not necessarily needed for safe frequency detection, to add filter stages to the upper two frequency bands to achieve a more pleasant and uniform sound experience in all frequency bands.

Input divider

A potential divider is used to limit the filter output to the maximum allowed input level of the amplifier module when the volume control of the module is set to maximum level. The filter stages are dimensioned in a way to get identical component values for all five dividers: $R1 / R2 = 22K / 3K3$.

It is important never to exceed the maximum input level of the amplifier, otherwise it would go into clipping which results in a square wave output on the loudspeaker. This would render the filter stage useless.

Amplifier

The amplifier used is a GF 1002 module. It comes with volume level control included on the module which allows for further adjustments of the tone level later during field tests.

The amplifier is a class D design which means the output to the loudspeaker is formed by a differential PWM signal with approx. 250 kHz base frequency. The sharp edges and the high currents generate significant EMI emissions; therefore, the amplifier shall be placed close to the loudspeaker to keep the connection wires as short as possible.

Loudspeaker

The loudspeaker must be able to produce a flat frequency response over the range of 160 Hz up to 6250 Hz.

If the existing speakers fit the requirement needs to be tested. The speakers are mounted in an open frame and are working more or less in an acoustic short cut situation. Especially for the two lower frequencies it must be tested if the sound pressure is sufficient.

Audio detection

Microphone module

The 2018 version uses a microphone module with a simple OpAmp pre amplifier. There is a high probability that such a simple module will be overloaded in a noisy environment. The pre amplifier will go into saturation and output the tone as a square wave. Detecting the fundamental becomes unreliable as described before.

Instead, the concept foresees a microphone module with automatic gain control. Common modules carry the MAX9814. The device features a low-noise preamplifier, variable gain amplifier (VGA), output amplifier, microphone-bias-voltage generator and AGC control circuitry. AGC gives a lot more headroom to possible clipping situations.

In addition, the gain can be selected manually to 40dB, 50dB or 60dB. On the workbench best results were achieved with gain limited to 40dB and A/R ratio set to 1:500. With A/R ratio 1:500 the gain recovers fastest after spikes. Must be double checked in field.

Audio analyzer

The audio analyzer circuit MSGEQ7 has already been used in the 2018 version of the 'Mobile', so no change here.

Theoretically a small computing module could be used to perform a simple Fourier analysis however, implementing and debugging the code adds an additional level of complexity which would increase development time. With the described countermeasures of filtering and AGC it should be possible to reliably detect the fundamental frequency with the audio analyzer module.

The STROBE and RESET inputs of the audio analyzer modules are all connected together to spare port resources on the hub. One cycle over all seven bands will read the outputs of all modules at the same time. The analog outputs are connected to five ADC inputs of the hub.

One property of the MSGEQ7 must be considered during programming: The internal scan rate is fast enough to take dedicated samples from the curve of the 160 Hz and 400 Hz signals. So, continuous readings with a certain amount of time in-between will show varying results depending on which part of the signal shape has been measured. This problem could be solved by reading four times in short sequence when a 160 Hz signal and two times in a sequence when a 400 Hz signal is expected. The results for the 160 Hz input always showed at least two 'hits' of high input levels and the 400 Hz at least one. (Will need some more experiments in the field test.)

Prototype boards

For first tests some prototype boards have been designed. Figure 4 - Figure 7 show the pinout.

Low Pass Filter

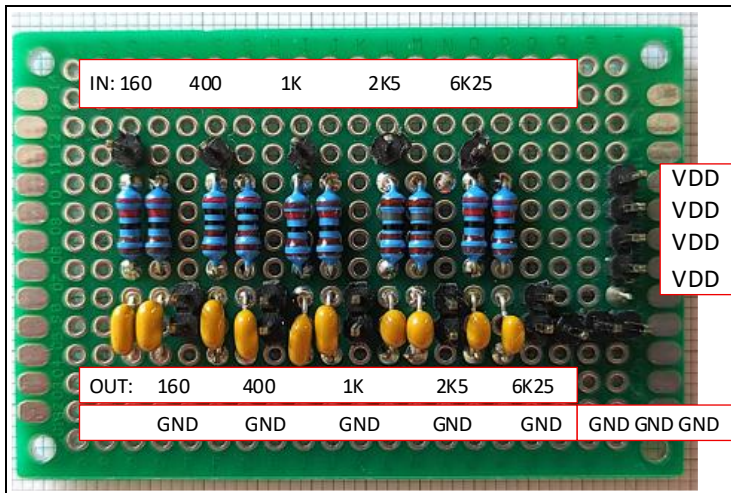


Figure 4: Filter module pinout

Microphone Module

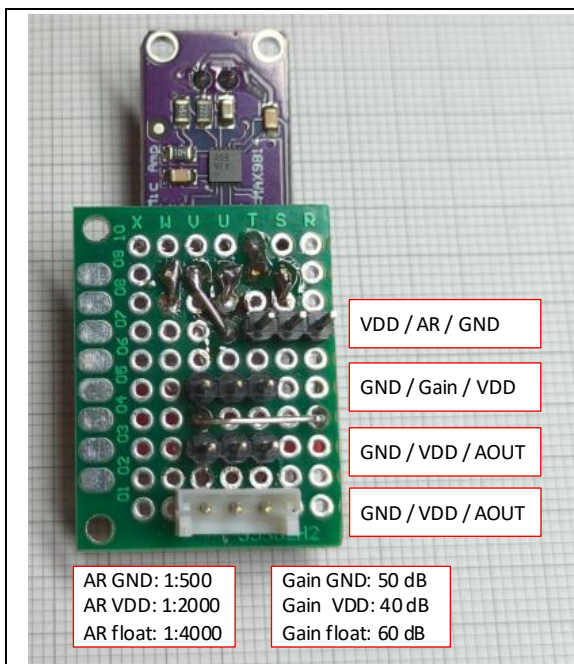


Figure 5: Microphone module pinout

Amplifier Module

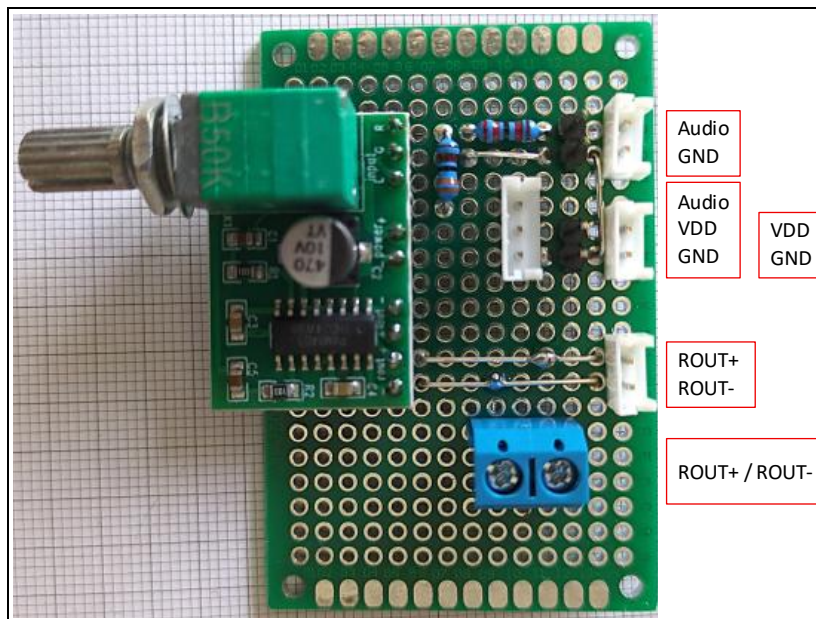


Figure 6: Amplifier module pinout

Audio Analyzer Array

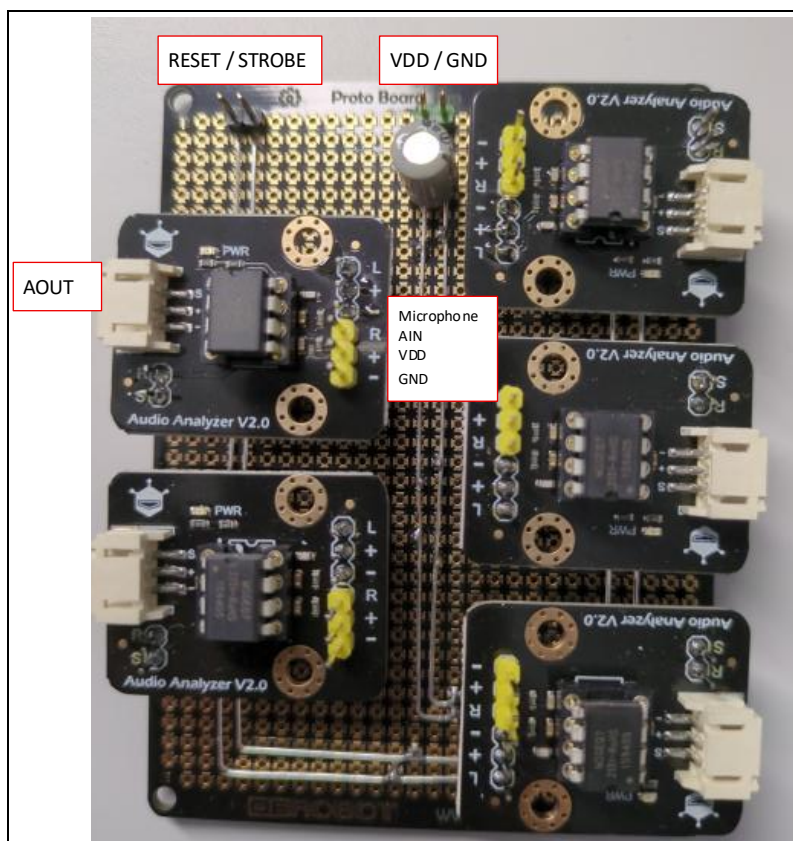


Figure 7: Audio analyzer array